Supplementary Material for "LF-NET: A Low-Frequency Self-Supervised Model for Coronary Artery Disease Detection from Phonocardiogram Signals"

Matthew Fynn, Milan Marocchi, Javed Rashid, Kayapanda Mandana, Yue Rong, and Goutam Saha

I. Quantitative Spectral Comparison of Pretraining Domains

To quantitatively compare the ocean pretraining corpus with PCG and an alternative physiological low-frequency source, ten 4 s fragments were sampled each from 296 ocean recordings, 103 PSG-Audio subjects, and 296 PCG subjects. All signals were analysed at 2 kHz over 20–1000 Hz, and statistics were computed at the independent recording-unit level.

Figure S.1 provides representative waveform and spectrogram examples from the three domains.

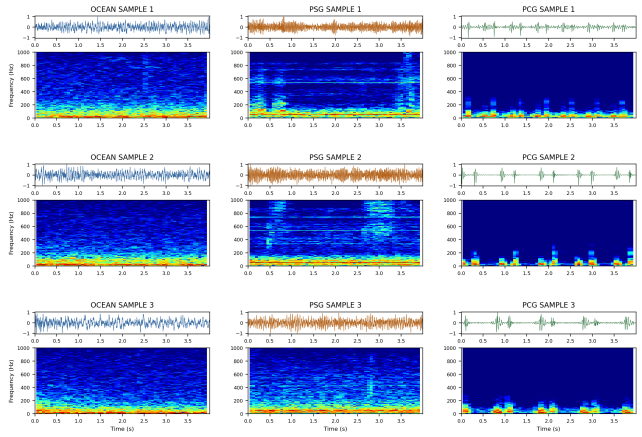


Fig. S.1. Representative waveform and spectrogram comparison of ocean, PSG-Audio, and PCG recordings.

For each short-time Fourier transform frame, the spectral centroid was calculated as

$$C(t) = \frac{\sum_{k\in\mathcal{B}} f_k |X_k(t)|^2}{\sum_{k\in\mathcal{B}} |X_k(t)|^2}, \tag{1}$$

where $\mathcal{B}$ contains frequency bins between 20 and 1000 Hz. Spectral bandwidth, the energy-weighted spread of the spectrum about its centroid; and 85% rolloff, the frequency below which 85% of the energy lies, were also calculated.

Spectral flux was used to quantify temporal variation in spectral shape. After ℓ_1 normalisation of each magnitude spectrum,

$$\tilde{X}_k(t) = \frac{|X_k(t)|}{\sum_{j\in\mathcal{B}} |X_j(t)|}, \tag{2}$$

the flux was calculated as

$$\Phi = \frac{1}{T-1}\sum_{t=2}^{T}\left[\sum_{k\in\mathcal{B}}\left(\tilde{X}_k(t) - \tilde{X}_k(t-1)\right)^2\right]^{1/2}. \tag{3}$$

To compare complete spectral distributions, Welch power spectral densities were normalised to unit area and treated as probability distributions over frequency. The Jensen–Shannon (JS) divergence between distributions P and Q was calculated as

$$\begin{aligned}\mathrm{JSD}(P \parallel Q) &= \frac{1}{2}D_{\mathrm{KL}}(P \parallel M) + \frac{1}{2}D_{\mathrm{KL}}(Q \parallel M),\\ M &= \frac{P+Q}{2}.\end{aligned} \tag{4}$$

where D_{KL} denotes the Kullback–Leibler divergence. Lower JS divergence indicates greater spectral similarity.

To estimate the expected within-domain divergence, the 296 PCG subjects were randomly divided into two equal subsets. One subset was used to form a PCG reference spectrum, while the remaining subjects were compared against this reference.

TABLE S.I
Spectral comparison of ocean, PSG-Audio, and PCG recordings. Values are mean ± standard deviation across independent recording units.

	Ocean	PSG-Audio	PCG
Spectral centroid (Hz)	100 ± 43	75.7 ± 18	42.1 ± 5.8
Spectral bandwidth (Hz)	132 ± 46	84.1 ± 22	31.4 ± 12
85% rolloff (Hz)	169 ± 110	101 ± 35	56.6 ± 12
Spectral flux	0.062 ± 0.010	0.070 ± 0.0051	0.107 ± 0.0086
JS divergence to PCG	0.159	0.283	0.038

Although PSG-Audio produced scalar spectral descriptors closer to PCG for several measures, the complete PSD distribution was more similar for ocean audio, with a JS divergence of 0.159 compared with 0.283 for PSG-Audio. Band-resolved analysis showed that ocean audio was particularly closer to PCG over 20–100 Hz, which contained 98.2% of the PCG spectral energy. These results provide quantitative support that the ocean corpus is well aligned with the dominant low-frequency content of PCG signals, while PSG-Audio provides an independent physiological low-frequency comparison.

II. LF-NET2D Architecture and Training Objective

LF-NET2D follows the masked-reconstruction self-supervised learning framework described for LF-NET in the main manuscript, but operates on two-dimensional MFCC representations. Each 4-s waveform (8000 samples at 2 kHz) was segmented into 40-ms frames with 50% overlap using centred analysis windows with zero-padding at the signal boundaries. The magnitude-squared Fourier spectrum was computed for each frame, followed by a 21-filter Mel filterbank and discrete cosine transform (DCT), yielding 20 MFCC coefficients. Each input fragment therefore produced a 20×201 MFCC matrix, with coefficients z-normalised across each time frame.

Figure S.2 shows representative MFCC representations of a PCG recording and an ocean sound recording used during self-supervised pretraining. Both exhibit energy concentrated within similar low-frequency regions, while the PCG representation retains its characteristic cardiac temporal structure.

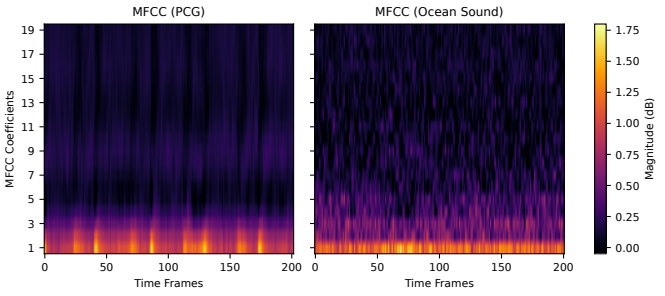


Fig. S.2. Representative MFCC matrices of a PCG signal (left) and an ocean sound recording (right).

A. LF-NET2D Architecture

The MFCC fragments were passed through a 2D U-Net with a backbone analogous to the 1D architecture presented in the main manuscript. The encoder contains four Conv2D blocks with channel widths of 16, 32, 64, and 256. The first two blocks use elongated 2×8 kernels followed by GELU activation and asymmetric MaxPool2D operations to primarily downsample the temporal dimension while preserving spectral resolution. The third and fourth blocks use 3×3 and 2×2 kernels, respectively, followed by a Conv2D$(256 \to 256, 4{\times}3)$ bottleneck.

The decoder mirrors the encoder resolution using ConvTranspose2D operations with channel transitions $256 \to 64$, $64 \to 32$, and $32 \to 16$. Unlike LF-NET, encoder-decoder skip connections are concatenative rather than additive to preserve distinct spectro-temporal information. Each upsampled representation is refined using a Conv2D layer with GELU activation, and a final $1{\times}1$ Conv2D produces the reconstructed MFCC map.

B. Masked MFCC Reconstruction Loss

Masking was initially performed in the time domain following the same procedure as LF-NET, with up to six masked frames. The binary time-domain mask was then mapped to the MFCC grid. Let

$$\mathbf{M}_{\mathrm{MFCC}} \in \{0,1\}^{F \times T_M}$$

denote the resulting mask, where F and T_M are the numbers of MFCC coefficients and time frames, respectively. The time-domain mask was averaged within each STFT window, and an MFCC frame was considered masked when more than 75% of its corresponding waveform samples were masked.

Given the reconstructed MFCC matrix $\hat{\mathbf{X}}$ and ground-truth matrix $\mathbf{X}$, the masked and unmasked reconstruction losses are

$$\mathcal{L}_{\mathrm{mask}} = \frac{\left\| \hat{\mathbf{X}} \odot (1 - \mathbf{M}_{\mathrm{MFCC}}) - \mathbf{X} \odot (1 - \mathbf{M}_{\mathrm{MFCC}}) \right\|_2^2}{\sum_{i,j} (1 - M_{\mathrm{MFCC},i,j})}, \tag{5}$$

and

$$\mathcal{L}_{\mathrm{unmask}} = \frac{\left\| \hat{\mathbf{X}} \odot \mathbf{M}_{\mathrm{MFCC}} - \mathbf{X} \odot \mathbf{M}_{\mathrm{MFCC}} \right\|_2^2}{\sum_{i,j} M_{\mathrm{MFCC},i,j}}. \tag{6}$$

These are combined as

$$\mathcal{L}_{\mathrm{MSE,2D}} = \mathcal{L}_{\mathrm{mask}} + 0.05\, \mathcal{L}_{\mathrm{unmask}}, \tag{7}$$

placing greater emphasis on reconstruction of the masked MFCC regions while maintaining consistency in the unmasked regions.

C. Encoder Consistency Loss

LF-NET2D applies a global encoder consistency objective between masked and unmasked views. Let

$$\mathbf{z}^{(m)}, \mathbf{z}^{(u)} \in \mathbb{R}^{B \times C \times H \times W},$$

denote the encoder outputs, where B, C, H, and W are the batch size, number of channels, height, and width, respectively. The spatial dimensions are flattened and ℓ_2-normalised to obtain representations $\hat{\mathbf{z}}^{(m)}$ and $\hat{\mathbf{z}}^{(u)}$. The encoder loss is

$$\mathcal{L}_{\mathrm{enc,2D}} = 1 - \frac{1}{BC} \sum_{b=1}^{B} \sum_{c=1}^{C} \left\langle \hat{\mathbf{z}}^{(m)}_{b,c,:}, \hat{\mathbf{z}}^{(u)}_{b,c,:} \right\rangle. \tag{8}$$

Unlike LF-NET, where encoder consistency is evaluated at temporal indices corresponding to masked and unmasked regions, LF-NET2D imposes global feature-level agreement across the spatial representation.

The final LF-NET2D objective is

$$\mathcal{L}_{\mathrm{2D}} = \alpha \mathcal{L}_{\mathrm{enc,2D}} + \beta \mathcal{L}_{\mathrm{MSE,2D}}. \tag{9}$$

LF-NET2D was trained using the Adam optimiser with a learning rate of 1×10^{-4}, batch size of 64, and 110 epochs, at which point the training loss had converged. Figure S.3 displays the loss curves.

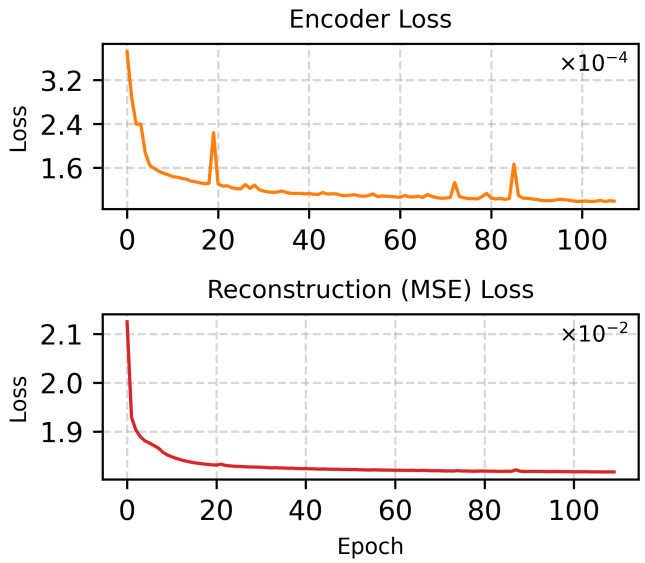


Fig. S.3. LF-NET2D pretraining loss curves.

III. Hybrid Model Hyperparameters

The hyperparameters selected for each single-channel hybrid model prior to channel-level fusion are reported in Table S.II. Here, n_{frag} denotes the number of input fragments used for each subject, and h_s denotes the size of the hidden fully connected layer. W2V, HuB, EnC, and OCE denote Wav2Vec 2.0, HuBERT, EnCodec, and OPERA-CE, respectively.

TABLE S.II
SELECTED SINGLE-CHANNEL HYPERPARAMETERS FOR THE HYBRID MODELS PRIOR TO MULTICHANNEL FUSION.

HF	LF	n_{frag} / h_s					
		Ch1	Ch2	Ch3	Ch4	Ch5	Ch6
W2V	LF-NET	91/1024	111/512	31/512	51/1024	31/512	111/256
W2V	LF-NET2D	111/1024	111/256	31/512	111/256	111/256	91/1024
HuB	LF-NET	91/1024	51/256	31/1024	51/256	31/512	31/1024
HuB	LF-NET2D	71/1024	31/512	51/256	111/512	91/1024	31/256
EnC	LF-NET	71/512	91/1024	31/256	31/256	31/512	111/256
EnC	LF-NET2D	31/1024	31/1024	71/256	91/512	91/256	111/512
OCE	LF-NET	91/512	51/1024	31/256	51/1024	91/1024	51/256
OCE	LF-NET2D	51/256	71/256	31/256	31/1024	111/1024	31/256

IV. Reciever-Operating Characteristics

Figure S.4 displays the receiver operating characteristic (ROC) curve for the highest-performing configuration (LF-NET+OPERA-CE), obtained using subject-level pseudo-probabilities (derived via softmax on the output layer) averaged across all cross-validation runs.

V. Random-Initialisation Ablation

To further assess the contribution of self-supervised pretraining, LF-NET was fine-tuned from random initialisation using the same downstream classification framework. Table S.III reports the subject-level performance across the evaluated multichannel combinations. Performance consistently

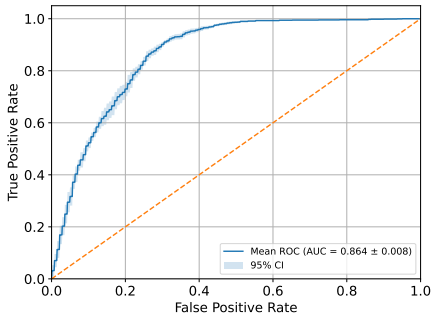


Fig. S.4. ROC-curve for LF-NET+OPERA-CE hybrid fusion model.

decreased without pretraining, with the best randomly initialised configuration achieving 76.82% accuracy compared with 79.80% for the pretrained LF-NET. These results support the contribution of the masked reconstruction pretraining stage to downstream CAD classification.

TABLE S.III
SUBJECT-LEVEL MULTICHANNEL PERFORMANCE OF RANDOMLY INITIALISED LF-NET.

Combination	ACC [%]	SENS [%]	SPEC [%]	MCC
3-4	73.72 (1.43)	79.62 (3.28)	67.29 (3.41)	0.48 (0.03)
3-4-6	75.68 (0.82)	83.92 (2.01)	66.74 (2.66)	0.52 (0.01)
2-3-4-6	76.35 (0.71)	84.05 (1.21)	68.03 (1.75)	0.54 (0.02)
1-2-3-4-6	76.62 (0.72)	83.75 (1.84)	68.86 (3.34)	0.54 (0.01)
1-2-3-4-5-6	76.82 (1.02)	83.49 (2.01)	69.57 (2.90)	0.54 (0.02)

VI. Noise-Removal Ablation

The noise-identification algorithm is presented in Algorithm S.1.

Algorithm S.1 Noisy Segment Identification

Require: Signal $x[0 \dots L-1]$, sampling rate f_s, frame length (s) T_f, threshold τ
Ensure: List of index intervals $\mathcal{I}$ containing frames flagged as noise

$N \leftarrow \lfloor L/(T_f \cdot f_s) \rfloor$ ▷ number of full frames
$F \leftarrow T_f \cdot f_s$ ▷ samples per frame
$\mathbf{E} \leftarrow \mathbf{0}_{1\times N}$ ▷ frame energies
for $i = 0$ to $N-1$ **do**
  $s \leftarrow iF; \quad e \leftarrow (i+1)F - 1$
  $\mathbf{E}[i] \leftarrow \sum_{n=s}^{e} x[n]^2$ ▷ sum of squares (energy)
$m \leftarrow \text{median}\big(\mathbf{E}[1 : N-2]\big)$ ▷ exclude first/last frame
$\mathcal{I} \leftarrow [\,]$ ▷ empty list of (start,end) indices
for $i = 0$ to $N-1$ **do**
  if $\mathbf{E}[i] > \tau \cdot m$ **then**
    $s \leftarrow iF; \quad e \leftarrow (i+1)F - 1$
    append (s, e) to $\mathcal{I}$
return $\mathcal{I}$

To further assess the contribution of the noise-removal procedure described in Algorithm S.1, we evaluated score-level fusion between the existing foundation-model encoders and LF-NET without noise removal. As shown in Table S.IV, performance generally decreased, with the largest deterioration observed in specificity. This indicates that artefact rejection is important for maintaining balanced classification performance under realistic clinical acquisition conditions.

When only the HM signal was used for noise identification, overall accuracy remained comparable to that obtained with

the full noise-removal procedure. However, the sensitivity-specificity balance deteriorated, with the gap increasing by approximately 8 percentage points. This suggests that incorporating both HM and NM information during artefact identification provides a more balanced downstream classification result.

TABLE S.IV
SUBJECT-LEVEL PERFORMANCE WITHOUT NOISE REMOVAL USING SCORE-LEVEL FUSION.

HF	LF	Combination	ACC [%]	SENS [%]	SPEC [%]	MCC
W2V	LF-NET	1-2-3-6	78.78 (2.49)	85.93 (3.39)	70.95 (2.63)	0.58 (0.05)
HuB	LF-NET	1-2-3-4-6	78.37 (1.44)	87.10 (3.18)	68.86 (1.17)	0.58 (0.03)
EnC	LF-NET	2-3-4-6	77.29 (0.44)	92.11 (1.65)	61.04 (1.51)	0.57 (0.01)
OCE	LF-NET	1-2-3-4-6	76.95 (2.41)	96.14 (1.60)	55.92 (4.81)	0.58 (0.04)

VII. PCG-ONLY SELF-SUPERVISED PRETRAINING

To further examine the SSL corpus ablation reported in the main manuscript, we analysed the training behaviour of LF-NET when pretrained using only the publicly available PCG corpus. This corpus comprised approximately 35,000 four-second PCG fragments, substantially fewer than the ocean-sound pretraining corpus.

Figure S.5 shows the MSE reconstruction loss during PCG-only self-supervised pretraining. Although the resulting model learned useful representations and achieved competitive downstream classification performance, the reconstruction loss exhibited greater variability during training than observed for the ocean-based pretraining configurations. This suggests that the substantially smaller PCG corpus may provide a less stable optimisation setting for masked-reconstruction pretraining.

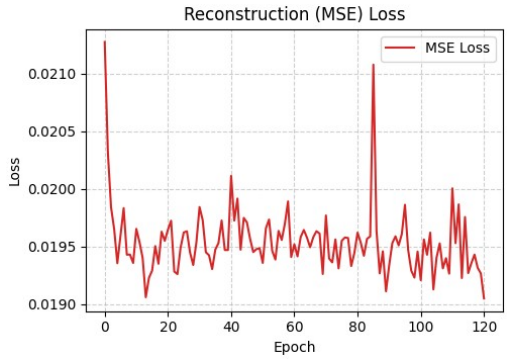


Fig. S.5. MSE reconstruction loss during PCG-only self-supervised pretraining of LF-NET.

Importantly, the PCG-only downstream results reported in the main manuscript demonstrate that useful representations can be learned directly from PCG data. However, the purpose of the ocean corpus is to provide substantially greater scale and diversity of low-frequency acoustic data, while limited PCG exposure introduces modality-specific cardiac structure. The complete downstream comparison between random initialisation, ocean-only, PCG-only, mixed ocean-PCG, LibriSpeech, and PSG-Audio pretraining is therefore retained in the main manuscript.